

Sales Performance Analysis

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Project Objective

- Clean and validate sales transaction data.
- Analyze sales performance using Excel.
- Create Pivot Tables for business analysis.
- Develop an interactive dashboard with KPIs and charts.
- Identify sales trends, customer behavior, and payment preferences.
- Generate actionable business insights for decision-making.

Project Description

This project focuses on analyzing a sales dataset containing 2,000 customer orders. The dataset was cleaned, transformed, and analyzed using Microsoft Excel. Pivot Tables, charts, KPIs, and dashboard visualizations were created to understand sales performance, customer purchasing patterns, product performance, and payment method trends.

Dataset Overview

The dataset contains:

- Order ID
- Customer Information
- Product Details
- Quantity Sold
- Unit Price
- Discount %
- Sales Amount
- Payment Type

Total Records Analyzed: **2,000 Orders**

Data Cleaning and Transformation

The following transformations were performed:

- Checked for duplicate records.
- Corrected data formatting issues.
- Handled missing values.

- Applied currency and percentage formatting.
- Created KPI calculations.
- Validated sales and discount values.
- Prepared data for Pivot Table analysis.

Descriptive Statistics

The dataset was analyzed using descriptive statistics:

- Mean
- Median
- Mode
- Standard Deviation
- Variance
- Maximum Sales Value
- Minimum Sales Value

DISCRIPTIVE STATISTICS FOR COST	
Mean	243.9706052
Standard Error	4.155279304
Median	240.63
Mode	245.9287654
Standard Deviation	131.335752
Sample Variance	17249.07975
Kurtosis	-0.952006166
Skewness	0.104451686
Range	478.41
Minimum	17.31
Maximum	495.72
Sum	243726.6346
Count	999
Confidence Level(95.0%	8.154086772

Summary

- Total Sales: ₹21,72,831.34
- Total Orders: 2,000
- Units Sold: 4,962
- Average Order Value: ₹1,086.42
- Average Discount: 15%
- Highest Order Value: ₹3,908.24
- Lowest Order Value: ₹18.46

Product_ID	Store_ID	Quantity	Unit_Price	Discount	Payment_Type	Total_Amount	Product name	COST	Loyalty Status	PROFIT	Total Revenue
PROD79	STORE9	4	₹ 202.59	6.00%	COD	₹ 761.74	Decorative Lamp Shade	₹ 112.42	Platinum	₹ 360.68	₹ 810.36
PROD33	STORE9	1	₹ 409.41	13.00%	PayPal	₹ 356.19	Modern Wall Clock	₹ 342.56	Gold	₹ 66.85	₹ 409.41
PROD43	STORE9	2	₹ 257.78	5.00%	Credit Card	₹ 489.78	USB External Hard Drive	₹ 21.34	Bronze	₹ 472.88	₹ 515.56
PROD80	STORE9	1	₹ 813.87	23.00%	Credit Card	₹ 626.68	Wall-Mounted Spice Rack	₹ 218.48	Bronze	₹ 595.39	₹ 813.87
PROD75	STORE9	4	₹ 255.48	3.00%	COD	₹ 991.26	Women's Ankle Boots	₹ 326.06	Silver	-₹ 282.32	₹ 1,021.92
PROD72	STORE9	1	₹ 724.74	15.00%	PayPal	₹ 616.03	Professional Jump Rope	₹ 43.29	Gold	₹ 681.45	₹ 724.74
PROD28	STORE9	2	₹ 67.55	3.00%	Credit Card	₹ 131.05	Sports Stopwatch Timer	₹ 17.31	Silver	₹ 100.48	₹ 135.10
PROD27	STORE9	2	₹ 717.21	15.00%	COD	₹ 1,219.26	Decorative Floor Vase	₹ 218.22	Platinum	₹ 997.98	₹ 1,434.42
PROD34	STORE9	2	₹ 166.26	14.00%	PayPal	₹ 285.97	Gaming Mechanical Keyboard	₹ 228.53	Silver	-₹ 124.54	₹ 332.52
PROD0	STORE9	2	₹ 540.48	29.00%	Credit Card	₹ 767.48	Brother Laser Printer	₹ 183.04	Gold	₹ 714.88	₹ 1,080.96
PROD66	STORE9	2	₹ 972.40	6.00%	COD	₹ 1,828.11	Women's Hair Dryer Brush	₹ 298.86	Gold	₹ 1,347.08	₹ 1,944.80
PROD99	STORE9	1	₹ 955.06	14.00%	Credit Card	₹ 821.35	Stainless Steel Cookware Set	₹ 449.40	Platinum	₹ 505.66	₹ 955.06
PROD57	STORE9	4	₹ 64.06	26.00%	Credit Card	₹ 189.62	Foam Roller	₹ 115.82	Platinum	-₹ 207.04	₹ 256.24
PROD46	STORE9	2	₹ 152.25	13.00%	Credit Card	₹ 264.92	Men's Winter Jacket	₹ 242.23	Silver	-₹ 179.96	₹ 304.50
PROD59	STORE9	2	₹ 107.97	6.00%	Credit Card	₹ 202.98	Men's Hair Styling Gel	₹ 48.01	Gold	₹ 119.92	₹ 215.94
PROD70	STORE9	4	₹ 930.38	0.00%	COD	₹ 3,721.52	Kitchen Storage Rack	₹ 465.34	Bronze	₹ 1,860.16	₹ 3,721.52
PROD94	STORE9	2	₹ 724.97	10.00%	Credit Card	₹ 1,304.95	Vitamin E Body Lotion	₹ 337.42	Platinum	₹ 775.10	₹ 1,449.94
PROD36	STORE9	2	₹ 117.34	9.00%	PayPal	₹ 213.56	Women's Summer Dress	₹ 144.53	Platinum	-₹ 54.38	₹ 234.68
PROD38	STORE9	1	₹ 40.25	10.00%	PayPal	₹ 36.23	Sports Water Hydration Pack	₹ 245.93	Platinum	-₹ 205.68	₹ 40.25
PROD0	STORE9	2	₹ 943.80	6.00%	Credit Card	₹ 1,774.34	Brother Laser Printer	₹ 21.34	Silver	₹ 1,844.92	₹ 1,887.60
PROD74	STORE9	3	₹ 627.29	27.00%	COD	₹ 1,373.77	Bluetooth Car Audio System	₹ 131.51	Platinum	₹ 1,487.34	₹ 1,881.87
PROD48	STORE9	2	₹ 636.51	25.00%	Credit Card	₹ 954.77	Car Dashboard Phone Holder	₹ 123.46	Silver	₹ 1,026.10	₹ 1,273.02
PROD71	STORE9	3	₹ 849.36	3.00%	Credit Card	₹ 2,471.64	Hair Repair Oil Treatment	₹ 484.62	Bronze	₹ 1,094.22	₹ 2,548.08
PROD4	STORE9	2	₹ 409.26	10.00%	PayPal	₹ 736.67	Portable Power Bank	₹ 245.93	Platinum	₹ 326.66	₹ 818.52
PROD89	STORE9	1	₹ 317.15	24.00%	COD	₹ 241.03	Face Foundation Kit	₹ 166.06	Bronze	₹ 151.09	₹ 317.15
PROD24	STORE9	1	₹ 882.73	10.00%	Credit Card	₹ 794.46	Foldable Camping Chair	₹ 174.02	Bronze	₹ 708.71	₹ 882.73
PROD81	STORE9	3	₹ 113.87	21.00%	COD	₹ 269.87	Badminton Racket Set	₹ 322.41	Bronze	-₹ 625.62	₹ 341.61
PROD26	STORE9	4	₹ 896.91	10.00%	Credit Card	₹ 3,228.88	Wireless Bluetooth Speaker	₹ 245.93	Gold	₹ 2,603.92	₹ 3,587.64

Pivot Table Analysis

Pivot 1: Store Wise Sales

- Identifies top-performing stores.
- Compares revenue contribution by store.

Pivot 2: Product Wise Sales

- Shows best-selling products.
- Highlights revenue contribution of each product.

Pivot 3: Payment Method Analysis

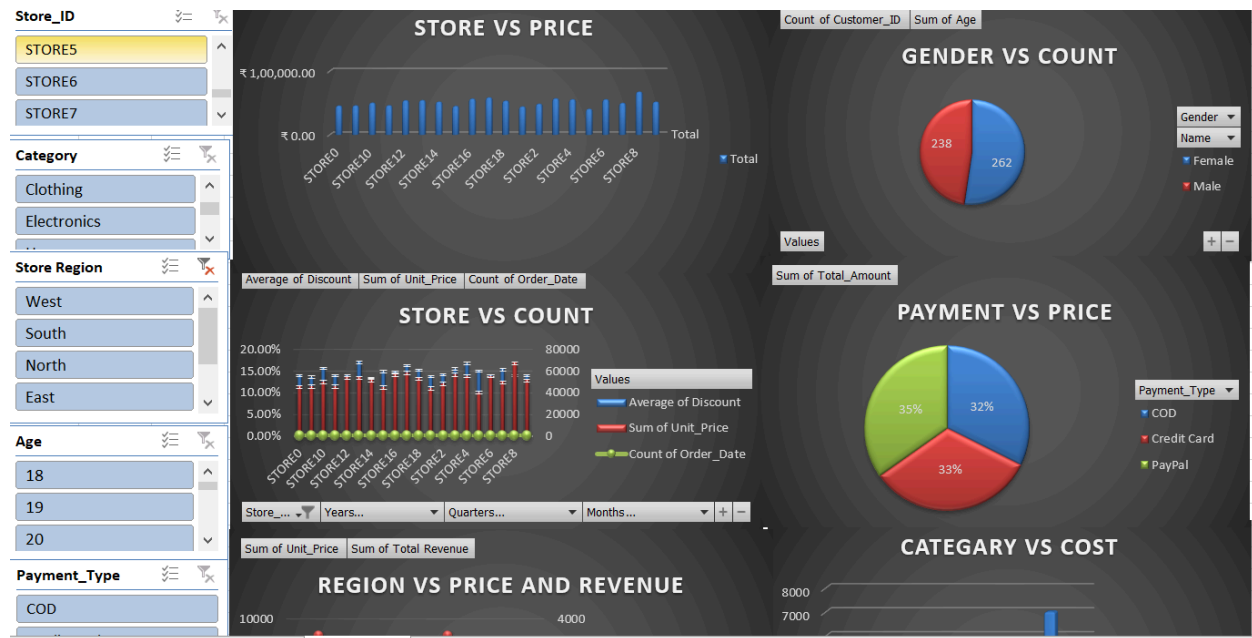
- Compares PayPal, Credit Card, Debit Card, and Cash payments.

Pivot 4: Customer Analysis

- Evaluates purchasing behavior and order frequency.

Pivot 5: Discount Analysis

- Measures the impact of discounts on sales performance.



Dashboard Components

The dashboard includes:

- KPI Cards
 - Total Sales
 - Total Orders
 - Average Order Value
 - Average Discount
- Sales by Store Chart
- Product Performance Chart
- Payment Method Analysis
- Customer Analysis
- Discount Impact Analysis
- Interactive Filters/Slicers

Key Insights

- Total sales exceeded ₹21 lakh.
 - Digital payment methods generated the highest transaction volume.
 - A small number of stores contributed the majority of sales revenue.
 - Several products consistently outperformed others in sales.
 - Discounts positively influenced customer purchases.
 - Customer demand remained consistent across the analyzed period.
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Business Questions Answered

Question	Answer
What is the total sales value?	₹21,72,831.4
How many orders were processed?	2,000
What is the average order value?	₹1,086.42
What is the highest order value?	₹3,908.24
What is the lowest order value?	₹18.46
What is the average discount given?	15%
How many units were sold?	4,962

Business Recommendations

- Increase stock availability for top-selling products.
 - Expand promotional activities in high-performing stores.
 - Improve marketing efforts for low-performing products.
 - Encourage digital payment adoption through incentives.
 - Strengthen customer retention programs.
 - Monitor discount strategies to maintain profitability.
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Conclusion

The Sales Performance Analysis Dashboard successfully transformed raw sales data into meaningful business insights through data cleaning, Pivot Tables, KPI analysis, and dashboard visualizations. The project helps management understand sales trends, customer behavior, and store performance, enabling better business decisions and future growth opportunities.